would face. But do not be mistaken: we can’t tackle all the possible issues here, and this doesn’t aim to be an exhaustive description of all the possible problems you would face, neither it is a runbook about how to operate your e-commerce application: only practice, rolling up your sleeves and getting burned while trying, can teach you that. What you can find in this chapter, besides a few examples of real-world technical challenges, is a hands-on example of the analysis process that brings you from a solution “on paper” for your use case, to a working application coping with the complex facets of reality.

11.1 An Application: Find Nearest Hub

Now that we have seen how to implement k-d trees (in chapter 9) and S-trees (chapter 10), we can focus on the problems we are able to solve by using these structures. And we can go back to our initial example: writing an application to find the closest warehouse/shop/retailer to customers, in real time. Figure 11.1 shows yet another variation of the map first introduced in figure 8.1, where we show a few (real) cities and some imaginary shops on the area around them. In our example, we need to find the closest shop able to deliver a certain item to customers, after they purchased that item on our website – for this reason we only show the kind of goods sold, for each shop, rather than its name or brand.